Minghao Mou

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Research Interests

I develop **learning-to-optimize** and **generative-modeling** based methods that reduce the online cost of repeatedly solving related optimization, equilibrium, and sampling problems: flow-matching models that learn *distributions over solutions*, amortized samplers for unnormalized distributions, and learned compressions of large-scale linear programs, each paired with theoretical guarantees and benchmarks against production solvers. Applications include robust learning, network flows, and large-scale power systems.

Learning to Optimize | Generative Models | Amortized Sampling | Distributionally Robust Optimization | Variational Inequalities

Education

Purdue University — West Lafayette, IN

Ph.D., Electrical Engineering (GPA 3.93/4.00) — 2023 – May 2028 (expected)

Advisor: Prof. Junjie Qin. Passed preliminary exam and advanced to Ph.D. candidacy, May 2026.

Coursework: Deep Learning, Generative Models, Optimization, Game Theory, Randomized Algorithms, Stochastic Processes.

The Chinese University of Hong Kong, Shenzhen — Shenzhen, China

B.S., Applied Mathematics, First Class Honours (GPA 3.80/4.00) — 2019 – 2023

Undergraduate research advisor: Prof. Shuang Li.

Coursework: Stochastic Simulation, Probability Theory, Data Structures & Algorithms.

Current Research: Learning to Optimize & Generative Models

Continual Flow Sampling for Distributionally Robust Optimization — Sep 2026 – Present

Generative models for robust learning · manuscript in preparation

- Showed that the worst-case laws of KL and entropic-Wasserstein/Sinkhorn DRO are energy-tilted densities indexed by nominal point x_0, and built a **single persistent, x_0-conditioned flow sampler** amortized across the training set and **warm-started across outer SGD steps**, extending data-free Flow Sampling to a moving, conditional target.
- Diagnosed a *discovery failure* of literal single-candidate self-replay (rollout directions statistically indistinguishable from random in pixel space) and designed **Best-of-N discovery + selected replay distillation**, a fixed, parallel budget of N forward passes with no iterative inner solver.
- On MNIST Sinkhorn-DRO adversarial training, reached **2.43% clean / 21.98% PGD-ℓ_∞ error with zero correction steps**, matching or beating a 30-step semi-amortized Langevin-correction baseline (2.44% / 22.6%) and cutting raw-sampler adversarial error from 25.1% to 10.4% vs. guided targets.
- Validated conditional amortization against closed-form and grid-quadrature ground truth on synthetic KL/Sinkhorn targets, and derived a first-order staleness bound guiding replay-buffer eviction.

Learning to Compress Linear Programs for Fast Value Evaluation — Sep 2026 – Present

Learning to optimize · manuscript in preparation

- Proposed learning a compact **dictionary of solutions** for large-scale LP families that share a polyhedral cone; a query is evaluated on the conic hull of stored solutions, which reduces to a **ray-ratio rule of two matrix–vector products** and yields a one-sided bound on the optimal value.
- Proved a ray-ratio theorem, a **stable sample-compression generalization bound** for value-approximation failures, and a critical-region-based result explaining when one stored ray covers many unseen queries.
- On SNDlib multicommodity-flow benchmarks, achieved **up to 1,088× online speedup over Gurobi** at a 0.1% relative-error target, faster than HiGHS basis reuse, PDLP, NVIDIA cuOpt, HPR-LP, and a learned GNN basis warm start; stress-tested under profile, capacity, and joint distribution shift.
- Scaled to a 2,000-bus Texas power-grid hosting-capacity panel: value-triggered repair raised median held-out coverage from 50.3% to **95.3%** with **∼3,400× median per-query speedup**.

Solving Parametric Variational Inequalities via Flow Matching Jul 2025 – Present

Generative learning to optimize · under review at ICLR 2027

- Built a **conditional flow-matching model that learns distributions over solutions** of parameterized variational inequalities, enabling amortized sampling with post-hoc best-of-N selection in place of per-instance iterative solves.
- Derived finite-sample guarantees linking flow-matching error to distributional, feasibility, and stationarity error of generated solutions.
- Evaluated on three non-monotone bimatrix-game families (up to 50 actions, 16 parameters; 12k training / 512 held-out instances, 5 seeds) and achieved **34–43× faster AC optimal power flow** queries than `pandapower`'s solver.

Other Research Experience

Strategic Storage Participation in Power Networks *@ Purdue* Feb 2026 – Present

- Modeled storage-aggregator / system-operator interaction as a Stackelberg game; solved equilibria via binding-pattern decomposition and an MIQP screen (IEEE 5–57-bus systems, 4.5–48 s per instance).

Braess' Paradoxes in Coupled Power and Transportation Systems *@ Purdue* Dec 2023 – Dec 2025

- Derived necessary and sufficient conditions for when capacity expansion in one network degrades both; designed detection algorithms and charging prices, tested on an IEEE 9-bus grid coupled to a Bay Area traffic network.

Nexus Cognizant Pricing of Workplace EV Charging *@ Purdue* Aug 2023 – Dec 2023

- Formulated bi-level pricing models that account for coupled power–transportation externalities (ACC 2024).

Latent Temporal Logic Point Processes *@ Research Assistant, CUHK-Shenzhen* Sep 2022 – May 2023

- Developed probabilistic event models with temporal-logic-rule priors to infer latent dynamics; implemented variational inference and Monte Carlo EM with sequential Monte Carlo for hidden-event inference.

Publications & Preprints

[1] **Minghao Mou**, Zaiwei Chen, Junjie Qin. "Solving Parametric Variational Inequalities via Flow Matching." *Under review, ICLR 2027.*

[2] Mehdi Davoudi, **Minghao Mou**, Junjie Qin. "Storage-Based Strategic Manipulation of Constraint-Binding Patterns in Power Networks." *arXiv:2609.15755, 2026.*

[3] **Minghao Mou**, Junjie Qin. "Braess' Paradoxes in Coupled Power and Transportation Systems." *arXiv:2512.12197, 2025.* [arXiv]

[4] Yaoyu Yang, Wei Wei, **Minghao Mou**. "Propelling DNA-Based Archival Storage With an Algorithmic Mindset." *IEEE Transactions on Molecular, Biological, and Multi-Scale Communications, 2026.*

[5] **Minghao Mou**, Sean Qian, Junjie Qin. "Nexus Cognizant Pricing of Workplace Electric Vehicle Charging." *American Control Conference (ACC), Toronto, 2024*, pp. 4275–4282. [IEEE]

Talks, Honors & Teaching

Talk: "Nexus Cognizant Pricing of Workplace Electric Vehicle Charging." ACC 2024, Toronto — Jul 2024

Talk: "Braess' Paradoxes in Coupled Power and Transportation Systems." INFORMS 2024, Seattle — Oct 2024

Honor: Academic Performance Scholarship, CUHK-Shenzhen — 2022

Teaching: Teaching Assistant, Random Variables and Signal Processing — 2026 Fall

Teaching: Teaching Assistant, Electric Machines — 2025 Fall

Teaching: Teaching Assistant, Functional Analysis & Partial Differential Equations, CUHK-Shenzhen — 2022 – 2023

Technical Skills

Machine learning: Flow matching, diffusion/flow samplers, amortized inference, distillation, robust training

Optimization: LP/conic duality, column generation, MIQP/MPEC, variational inequalities, equilibria

Software: Python, PyTorch, NumPy, SciPy, pandas, Gurobi, HiGHS, PDLP, cuOpt, MATLAB, SQL, Git